Profile AWS Benchmark 2024

FIS Profile V7.6.8 / GT.M V7.1-003

RedHat Enterprise Linux 9.4

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Performance Testing Objectives

- Demonstrate Performance and Scalability in the Public Cloud
- Conduct Profile Core Performance Tests on AWS EC2 Instance Types
- Analyze Profile V7.6.8 / GT.M 7.1-003 Performance
  - Deploy and Test High End Provisioned IOPS SSD Storage
  - Staged Profile Performance Configurations and Test Cases
    - Buildout of 5M, 10M and 25M Customer/Account Databases
    - Testing Includes Online Simulation and Key Overnight Batch Jobs
    - Encrypted Database
    - Spanned Global Regions
    - Online GT.M Database Backup
    - Increased Online Virtual Users and Batch Thread Analysis
  - Validation of the Profile/RHEL 9 Operational Readiness Checklist
  - Document Current Results (Performance Summary and Benchmark Report)

Performance Overall Summary

- Deploying FIS Profile in the public cloud provides robust performance and scalability, making it a viable option for a supported production architecture.
- Achieved 8,000+ TPS for Online Processing using larger AWS Instance Type with average response times of 6 milliseconds.
- Increased AWS Instance Type Sizes with additional CPU and memory resources allows for further online throughput and better batch runtime performance.
- Larger AWS Instance Types allow higher maximum IOPS and throughput, improving both online and batch performance.
- Upgrading from General Purpose SSD (gp3) to Provisioned IOPS SSD (io1) allows for increased IOPS and throughput improving both online and batch performance.
- Improved disk storage layout, including multiple striped disks per volume group and filesystem, increases the overall online throughput and batch runtime performance.
- Database Encryption Overhead observed at 3%-5% as Compared to a Standard Non-Encrypted Database.
- Preferred Linux Tuned-ADM setting is set to latency performance.

AWS EC2/RHEL 9.4 Configurations

Amazon EC2 M-series instances are general-purpose instances that provide a balance of compute, memory, and networking resources. They are suitable for a wide range of applications, including web servers, application servers, small and mid-sized databases.

M5.2xlarge

- 4 X 2.5GHz Intel Xeon Platinum 8175M CPUs
- Hyperthreads = 2
- 8 vCPUs
- 32 GB Memory
- RHEL 9.4

M5.4xlarge

- 8 X 2.5GHz Intel Xeon Platinum 8175M CPUs
- Hyperthreads = 2
- 16 vCPUs
- 64 GB Memory
- RHEL 9.4

M5.8xlarge

- 16 X 2.8GHz Intel Xeon Platinum 8259CL CPUs
- Hyperthreads = 2
- 32 vCPUs
- 128 GB Memory
- RHEL 9.4

M5.24xlarge

- 48 X 2.8GHz Intel Xeon Platinum 8259CL CPUs
- Hyperthreads = 2
- 96 vCPUs
- 384 GB Memory
- RHEL 9.4

AWS EC2 Storage/Disk Configuration

General Purpose SSD (gp3) Storage

- 3000 Max IOPS/125 MBps Throughput

XFS File System Layout

/profile	1200 GB
/gbls	500 GB
/journals	500 GB
/backup	500 GB

Provisioned IOPS SSD (io1) Storage

- 18750 Max IOPS/250 MBps Throughput

XFS File System Layout

/profile_gbls1	200 GB
/profile_gbls2	200 GB
/profile_gbls3	200 GB
/profile_gbls4	200 GB
/profile_gbls5	200 GB
/profile_gbls6	250 GB
/profile_gbls7	250 GB
/profile_gbls8	250 GB
/profile_gbls9	250 GB
/profile_journals1	120 GB
/profile_journals2	120 GB
/profile_journals3	120 GB
/profile_journals4	120 GB
/profile_journals5	120 GB
/profile_journals6	120 GB
/profile_spool	50 GB
/profile_backup	1000 GB

Profile Configuration

- Profile V7.6.8
  - Profile Application: master_v76_6241_42ef7
  - Profile Framework: master_v40_5388_3f66
- GT.M V7.1-003
- Customer and Account Database Sizes

Class	Type	5M Database	10M Database	25M Database
CIF	Personal	5,000,000	10,000,000	25,000,000

Class	Group	Type	5M Database	10M Database	25M Database
Deposit	SAV	300	1,000,000	2,000,000	5,000,000
Deposit	CD	350	500,000	1,000,000	2,500,000
Deposit	DDA	400	2,000,000	4,000,000	10,000,000
Loan	LN	500	500,000	1,000,000	2,500,000
Loan	RC	600	500,000	1,000,000	2,500,000
Loan	MTG	700	500,000	1,000,000	2,500,000
		Total	5,000,000	10,000,000	25,000,000

Online and Batch Workloads

- Online Workload
  - 50% Financial Transactions (DDA and SAV deposit/withdrawals TSSP)
  - 50% Inquiry Messages (Customer and Account Inquiry MSQL)
  - Processed 1,000,000+ online transactions per test

Type	Service Class	Message Percentage
DDA Deposit	TSSP	15%
DDA Withdrawal	TSSP	15%
SAV Deposit	TSSP	10%
SAV Withdrawal	TSSP	10%
Customer Inquiry	MSQL	25%
Account Inquiry	MSQL	25%

- Batch Processing
  - Financial Batch Posting (@QUE066)
  - Interest Posting (@QUE055)
  - Accrual Processing (@QUE039)

Function	Description	Record Count 5M Database	Record Count 10M Database	Record Count 25M Database
@QUE066	Batch Posting	500,000	1,000,000	2,500,000
@QUE055	Interest Posting	150,000	300,000	1,500,000
@QUE039	Accrual Processing	5,000,000	10,000,000	25,000,000

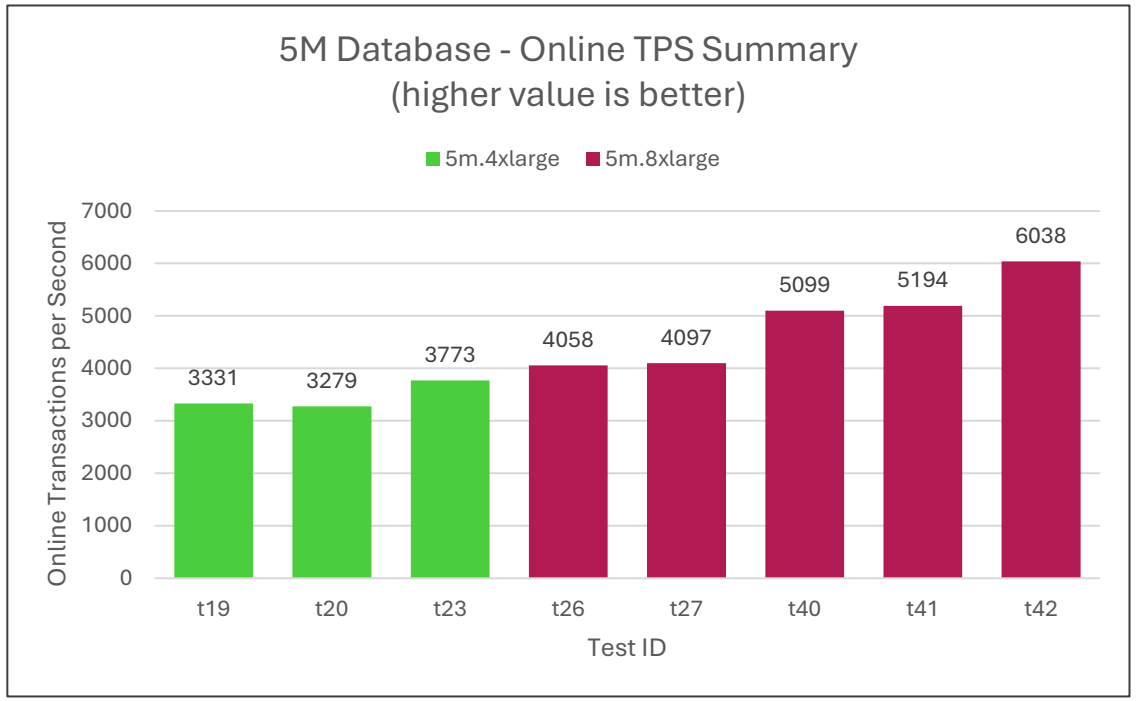
Note – running key critical path batch jobs to evaluate Profile Performance

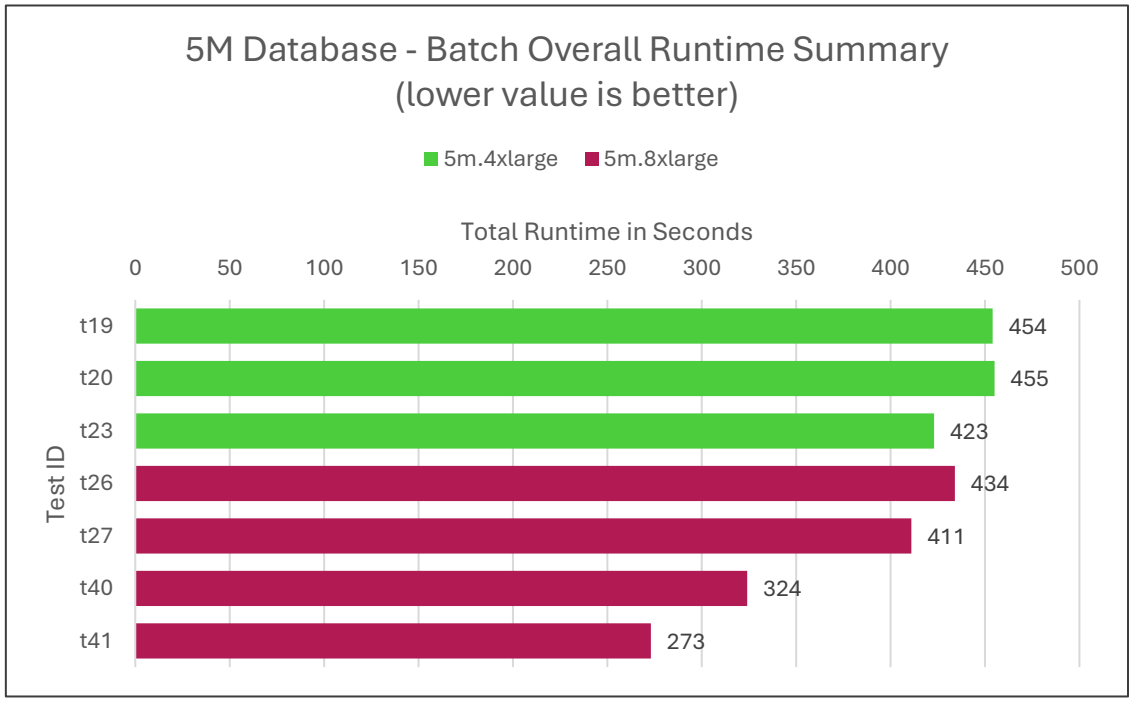
Profile Summary Results

5M Database Size

As test IDs increase, using larger instance types and adding processing threads.

- Increased Virtual Users for Online Simulation
- Increased Thread Count for batch processing



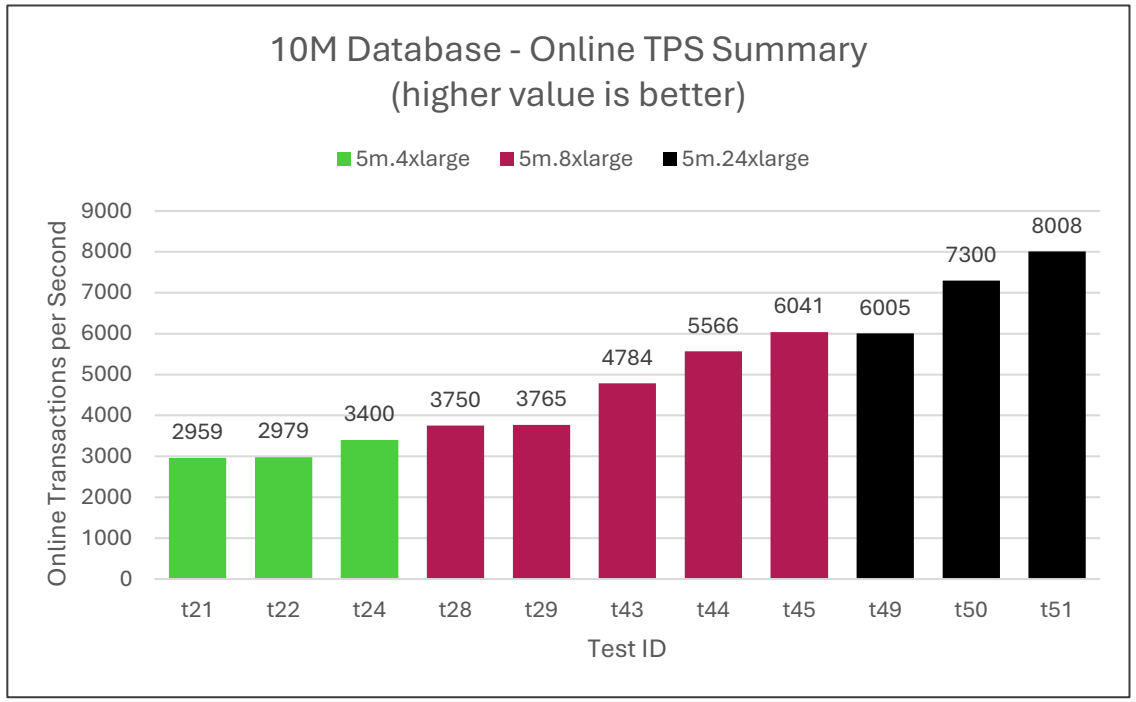


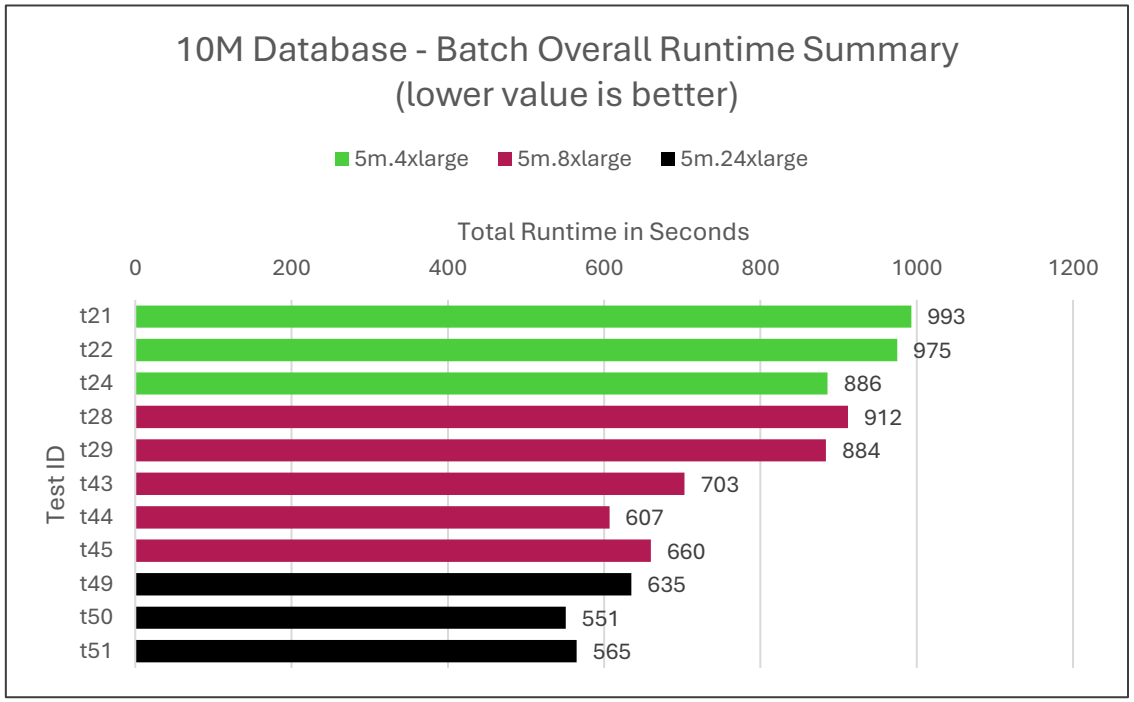
Profile Summary Results

10M Database Size

As test IDs increase , using larger instance types and adding processing threads.

- Increased Virtual Users for Online Simulation
- Increased Thread Count for batch processing





Profile Summary Results

25M Database Size

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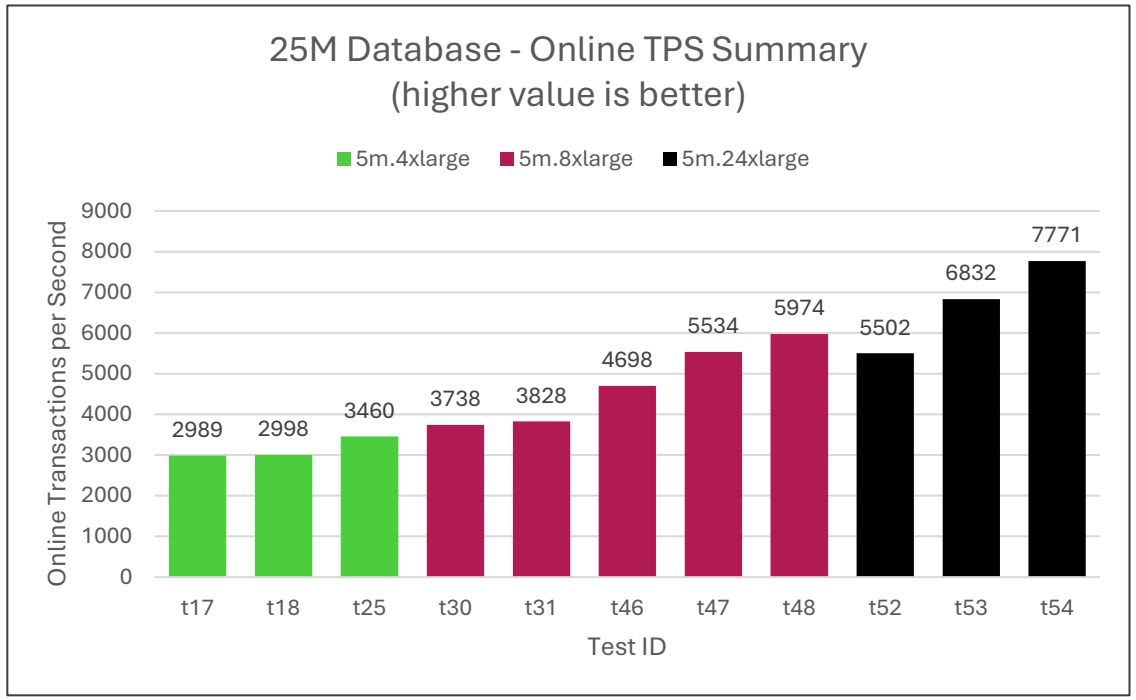
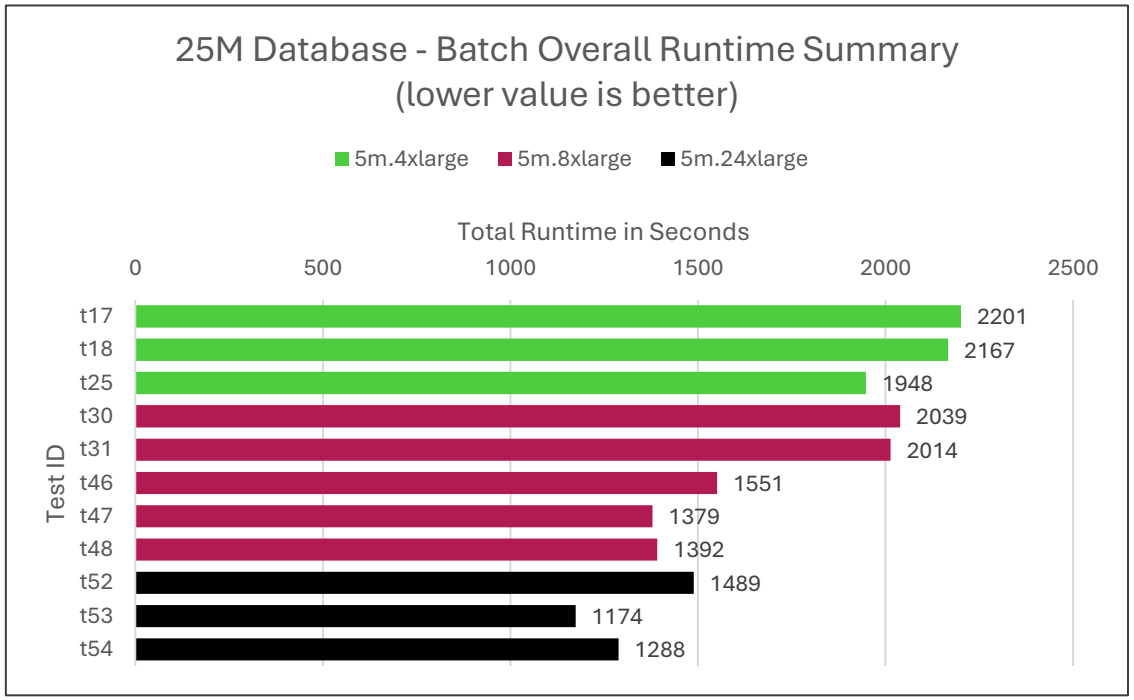
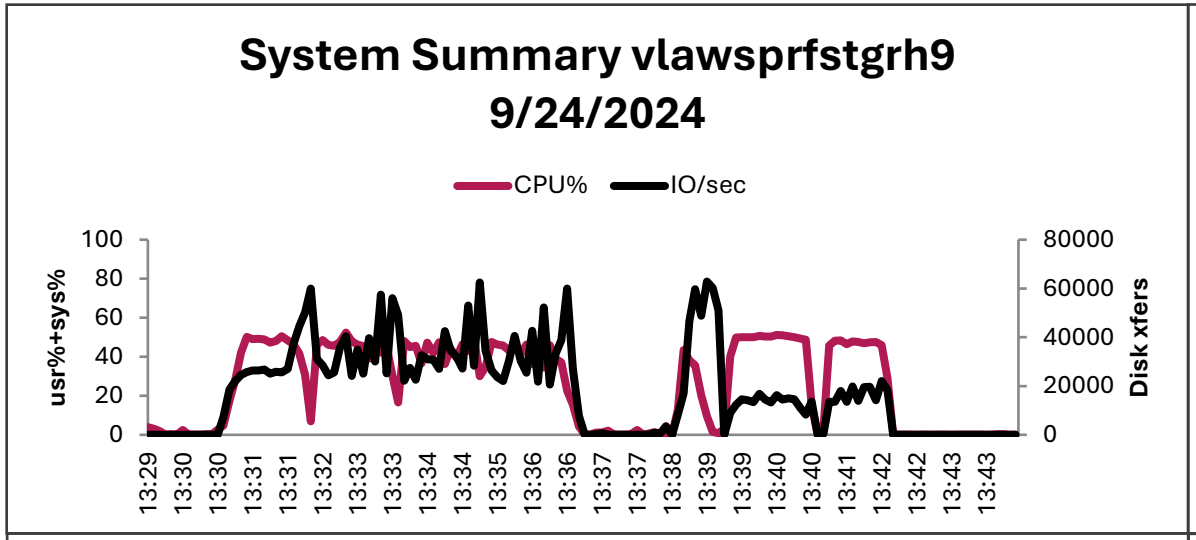
- Increased Thread Count for batch processing

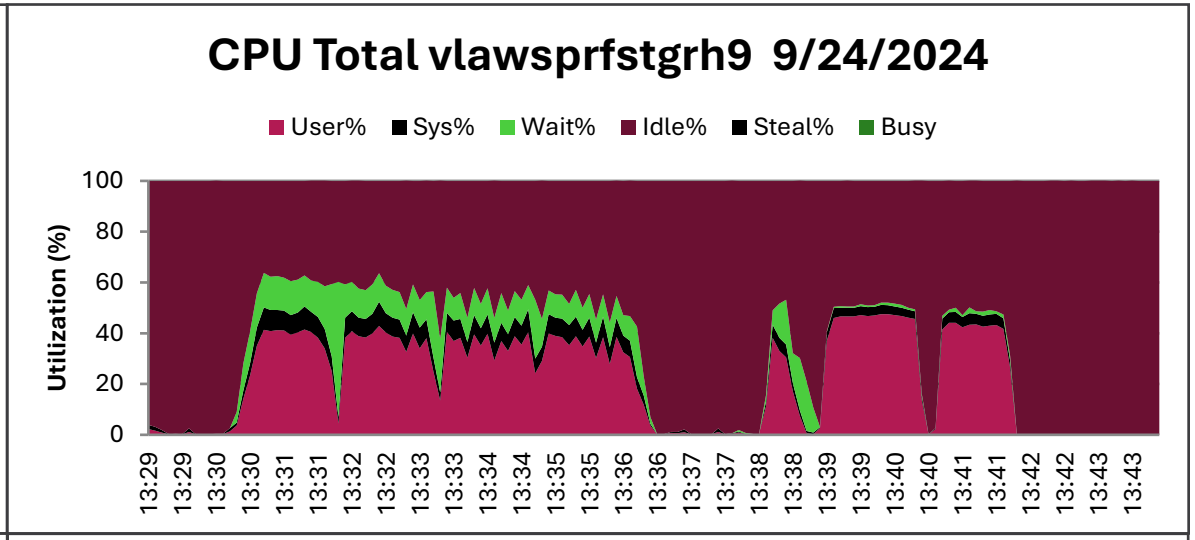


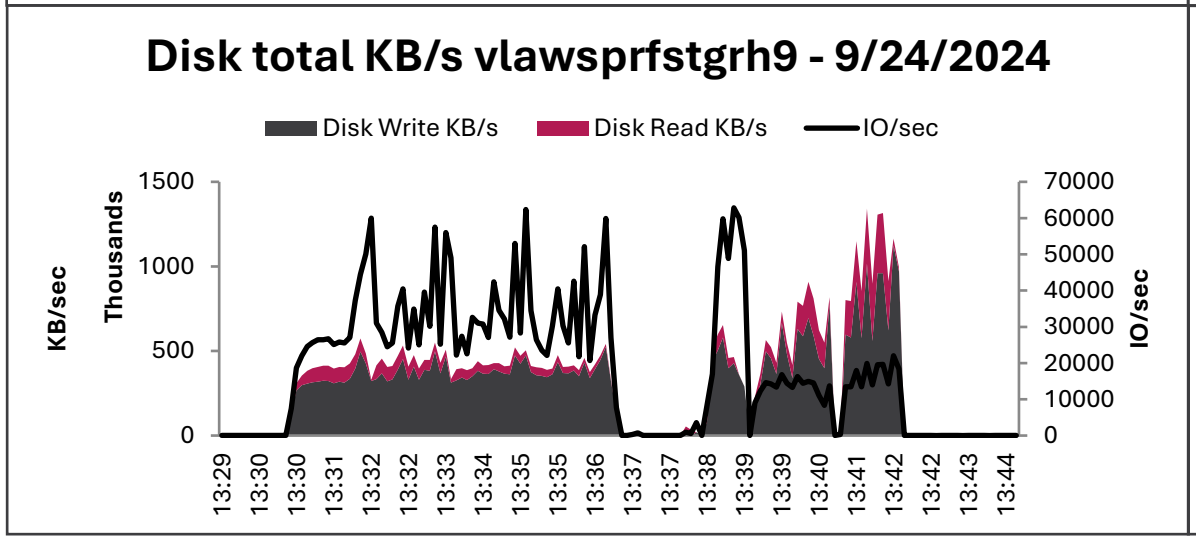


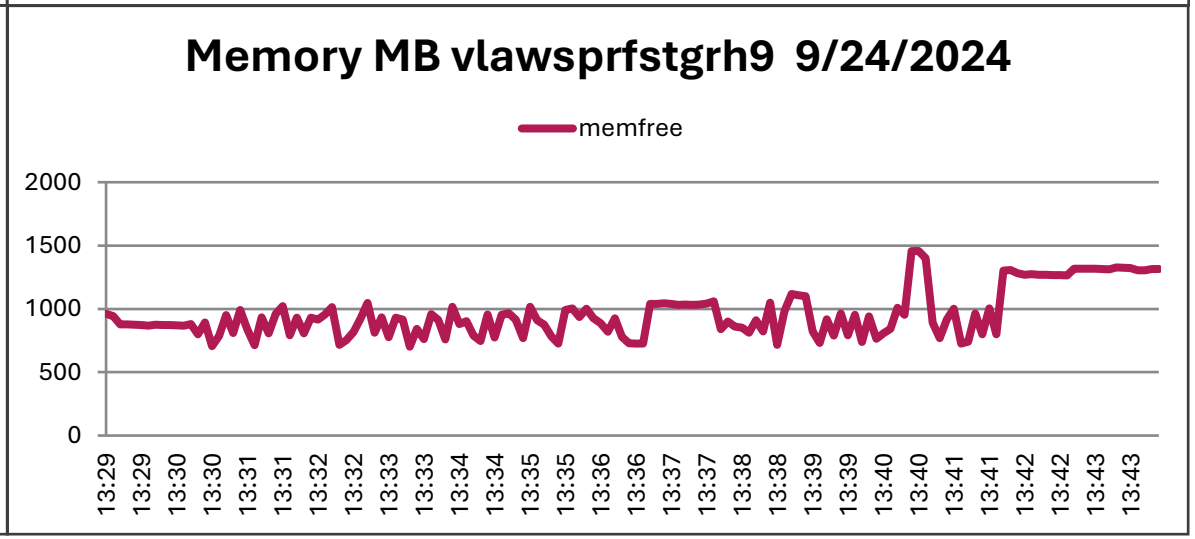
Host Server Utilization Statistics (example)

5M Database Size – t41 M5.8xlarge



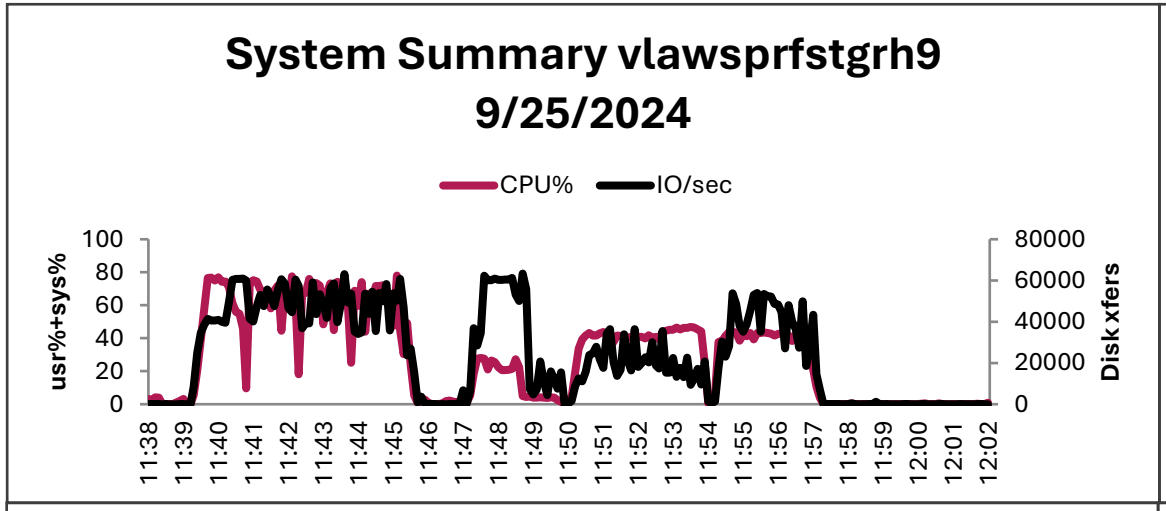


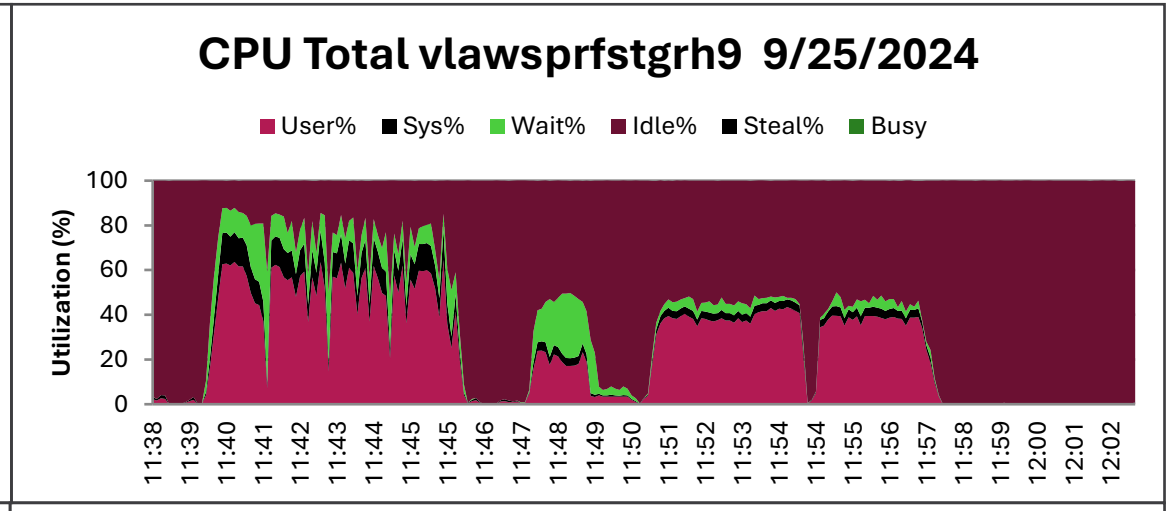


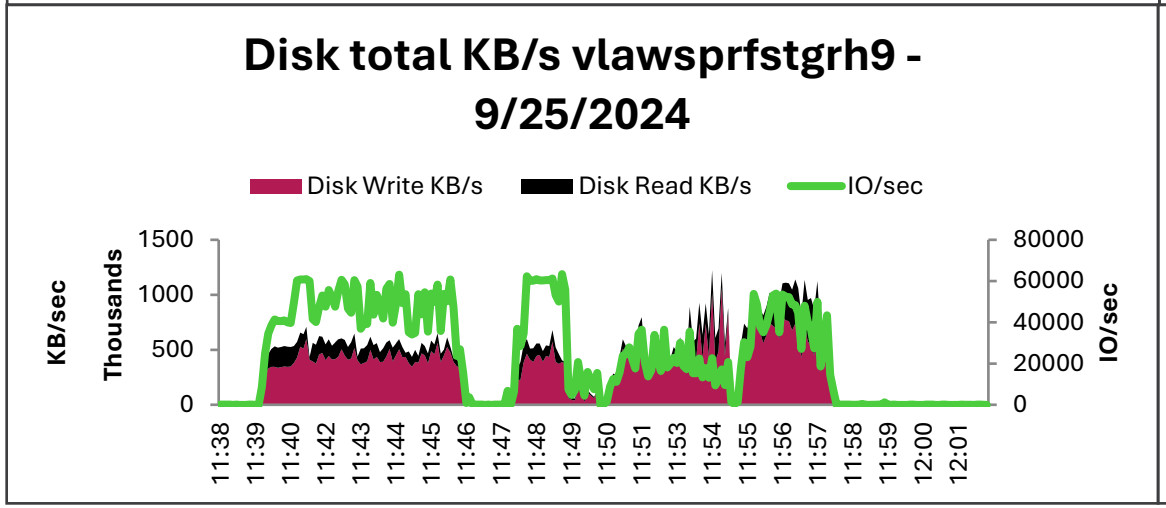


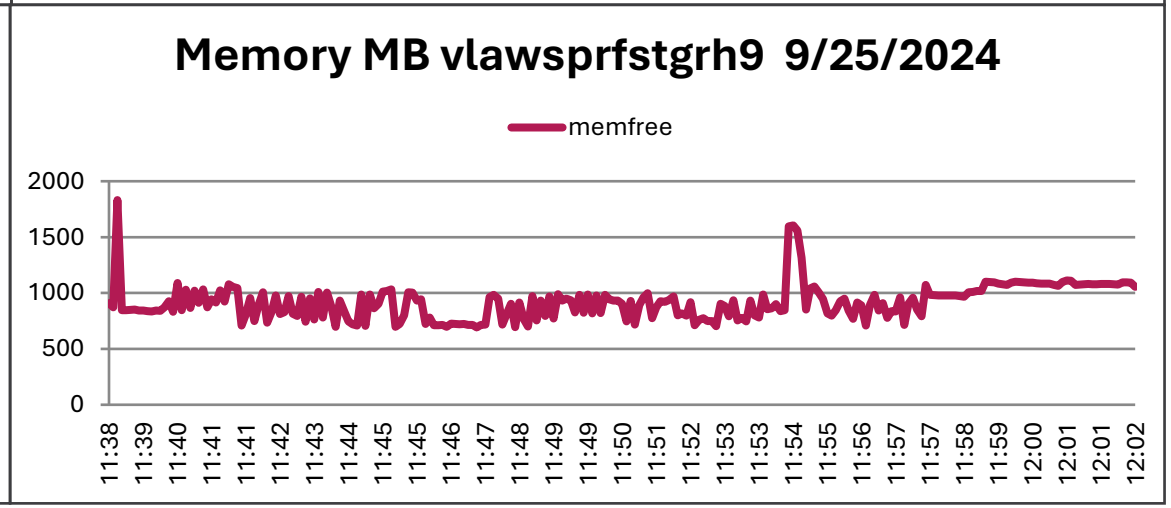
Host Server Utilization Statistics (example)

10M Database Size – t45 M5.8xlarge



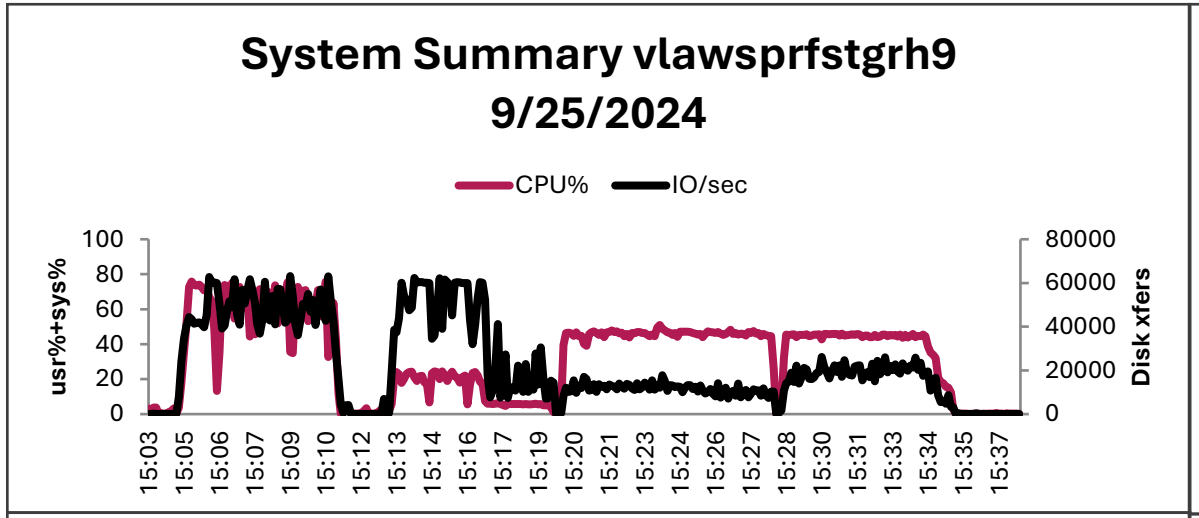


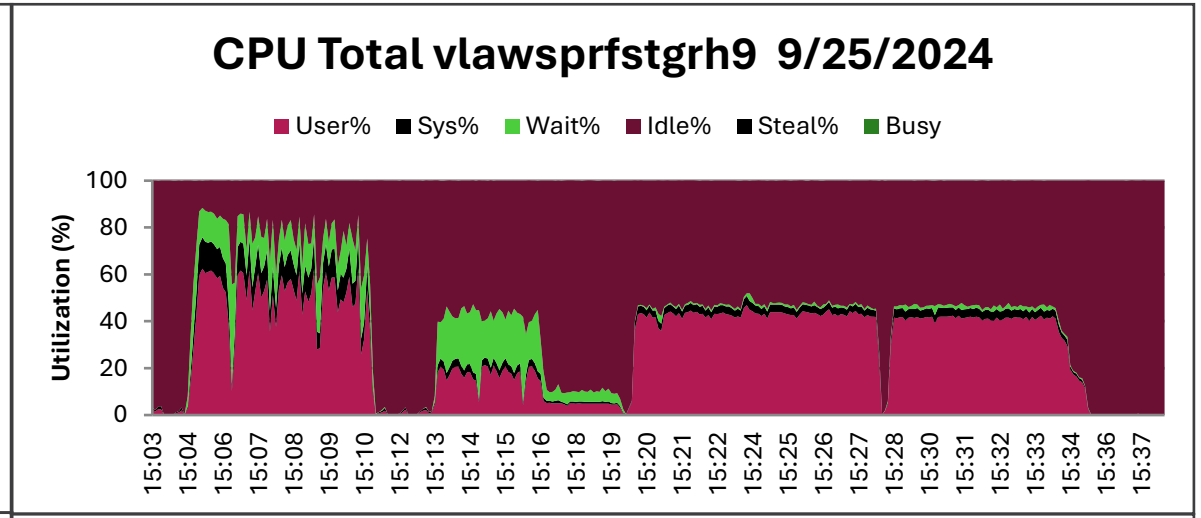


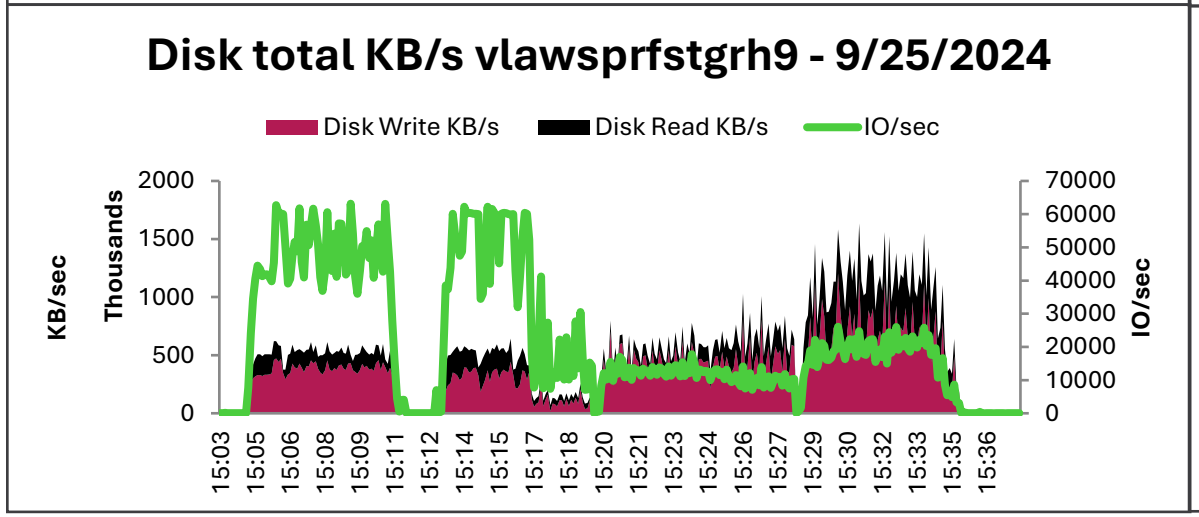


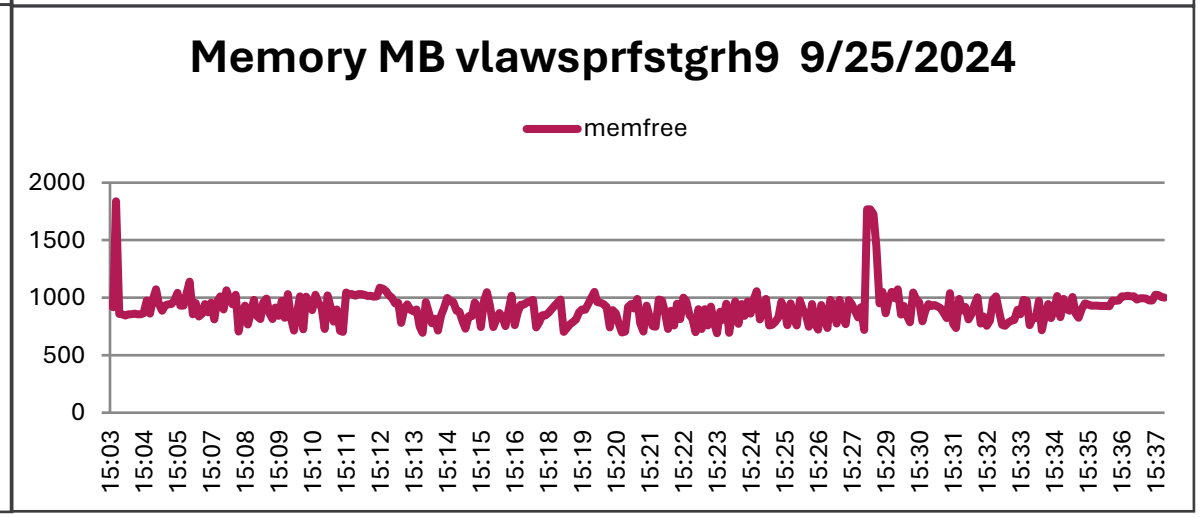
Host Server Utilization Statistics (example)

25M Database Size – t48 M5.8xlarge



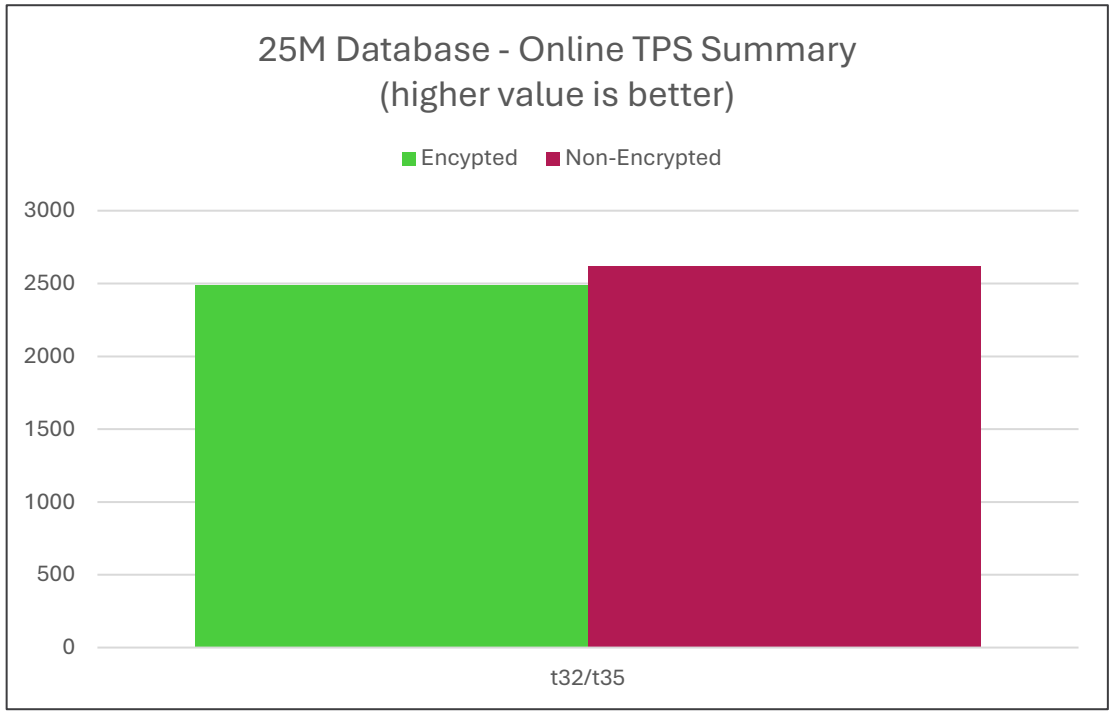


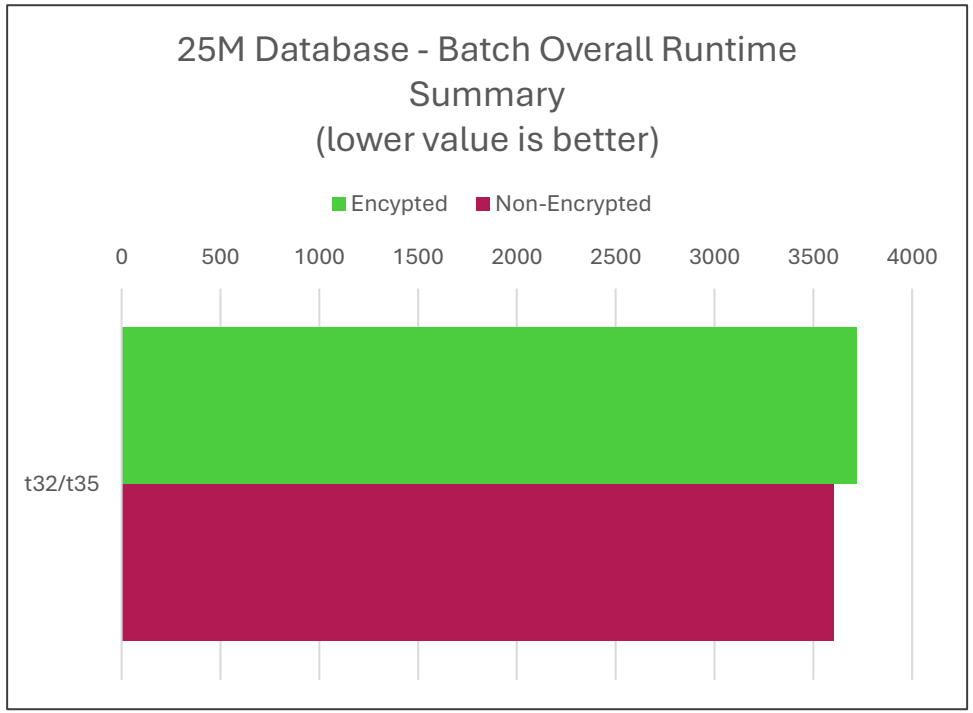




Database Encryption Findings

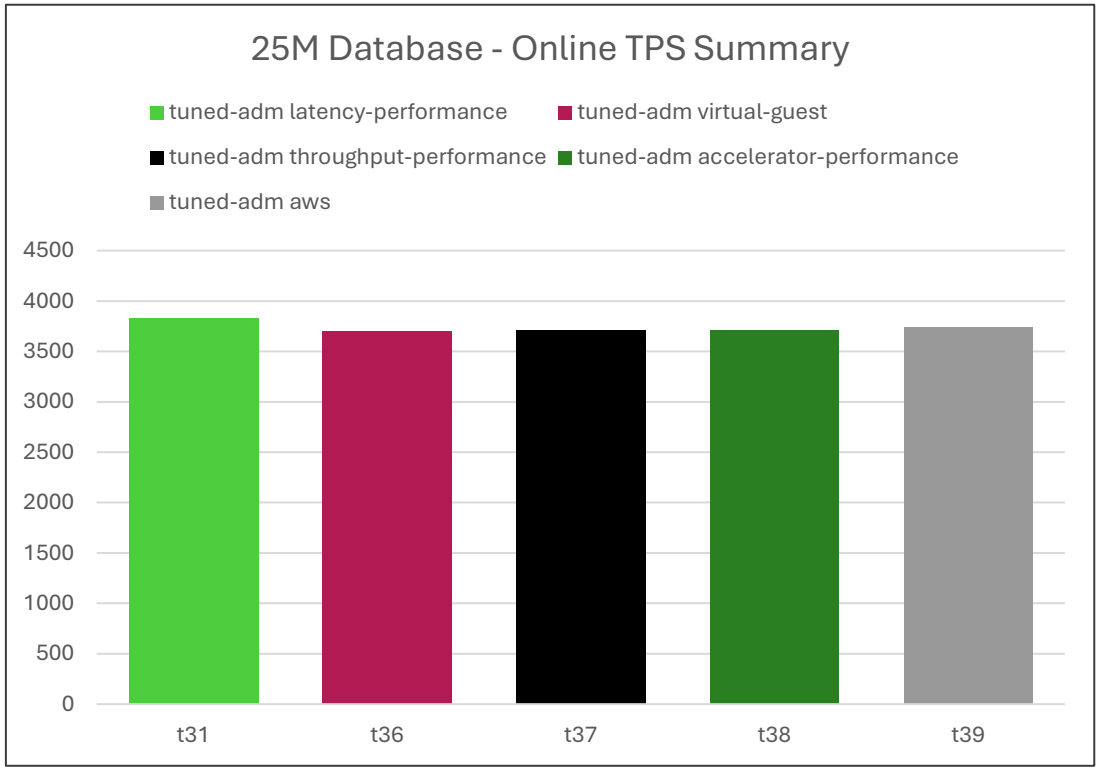
- Encryption Enabled Impacts (AES 256 Plug-in Library Architecture)
  - Online showed average of ~5% decrease in online TPS.
  - Batch processing showed average of ~3% Decrease in runtime performance.

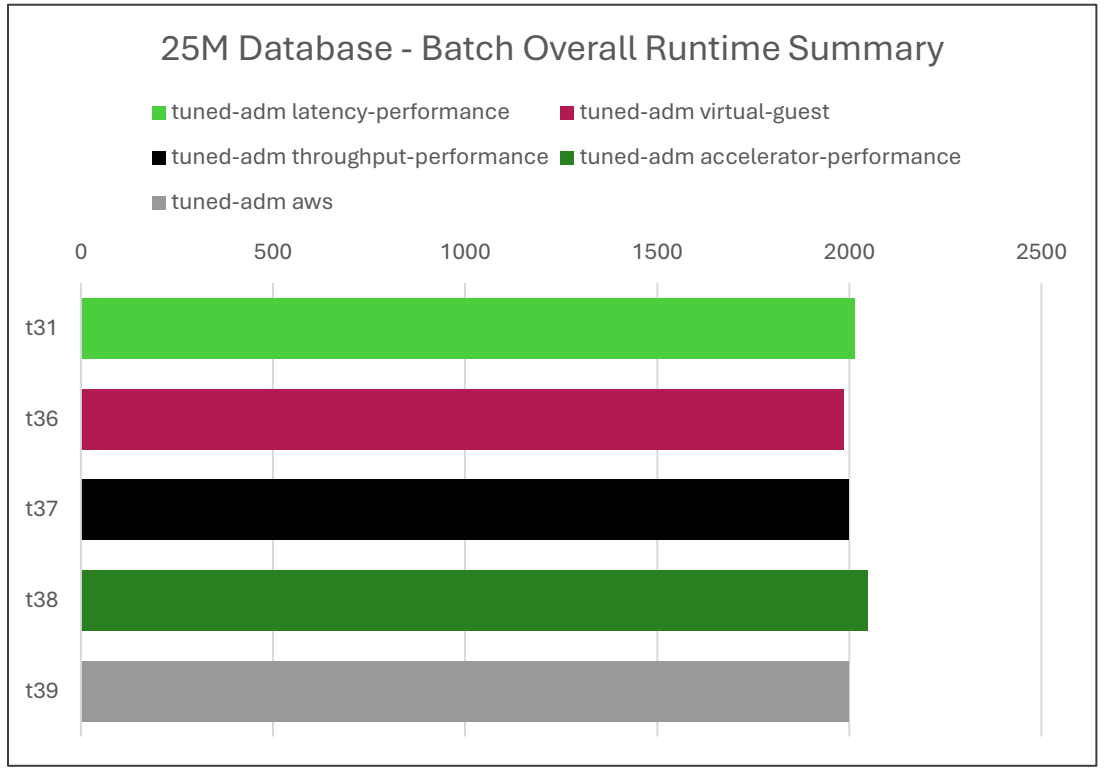




Tuned-ADM Findings

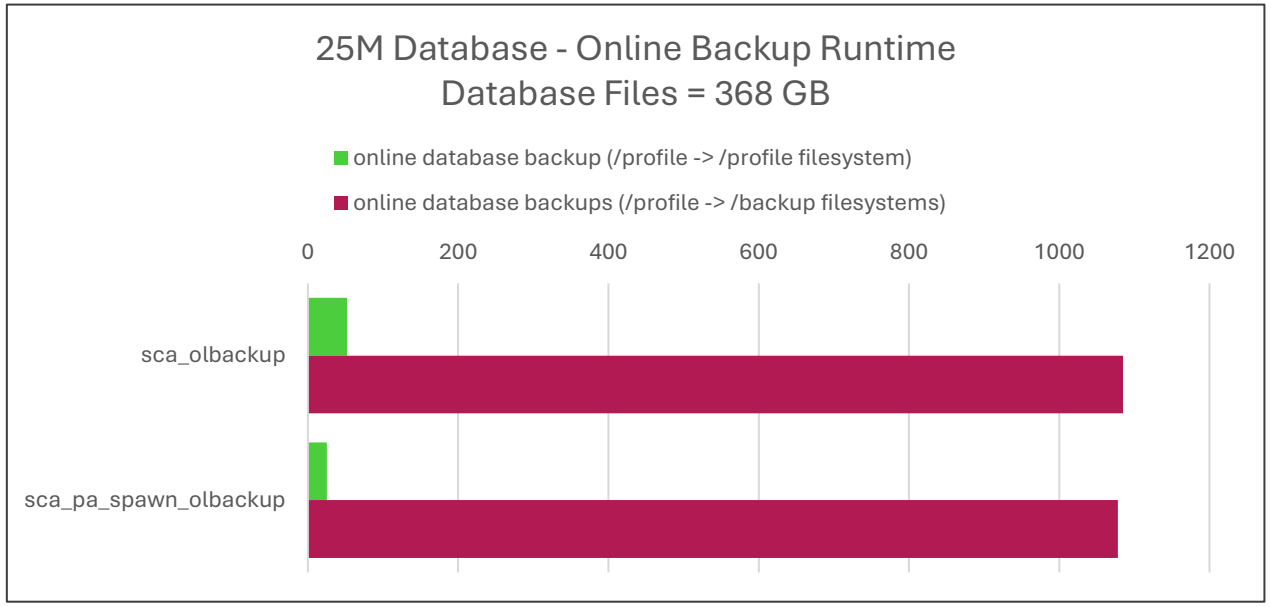
- tuned-adm best practice for Profile
  - current profile=latency-performance





Online Backup Summary

- sca_olbackup.sh – backup all regions of a GT.M database
- sca_pa_spawn_olbackup.sh – multi-threaded online backup
- GT.M mupip utility backup
- Significant Reduction when copying files within the same filesystem using RHEL 9 (reference links).



Next Steps / Future Performance Testing

- Finalize Detailed Benchmark Report
- AWS Profile Replication Testing
  - Replication timings and backlog levels
  - Local secondary instances setup
  - DR secondary instances setup
- MTM-SSL Enabled Performance Characterization
- AWS Console Based Monitoring Tools Review
- Testing and Characterization of EC2 Storage Optimized Instances
- Testing and Characterization of io2 Disk Types (higher durability and performance than io1)
- Evaluate Reflinks (reference links) for Profile backups

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